

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: pC59, an E2 of BPV-1 expression plasmid; p407, an E2 responsive CAT plasmid.
Species Used	Mouse, C127, fibroblast, mammary tumor		

Before the Pulse

Cell Growth Medium	DMEM + 10% Fetal Bovine Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Not given
		Pre-pulse Incubation	Room temperature
Wash Solution	1x Phosphate Buffered Saline, 3 times		

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature		
Electroporation Medium*	DMEM + 10% FBS + 5mM BES, pH 7.2, 50 µg salmon sperm, carrier DNA	Cuvette Gap	0.4 cm
Cell Density	1 to 5 x 10 ⁶ cells / ml	Voltage	0.21 kV
Volume of Cells	0.5 ml	Field Strength	0.525 kV/cm
DNA Concentration	5 to 10 µg DNA		
DNA Resuspension Buffer	in 5 to 10 µl 10 mM Tris Buffer, pH 8.0	Capacitor	960 µF
Volume of DNA	5 to 10 µl	Resistor	(Pulse Controller) Ω none
After the Pulse		Time Constant	20 msec
Outgrowth Medium	DMEM, 10% Fetal Bovine Serum		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Similar protocol has been published by another lab. Reference is given below:

Ustav, M. and Stenlund, A. (1991) *EMBO J.* **10**(2): 449-457.

Outgrowth Temperature	37 °C
Length of Incubation	48 hours
Selection Method or Assay Used	CAT assay

Electroporation Efficiency	Not given
Per Cent Survival	40 to 60%

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